Water consumption

A

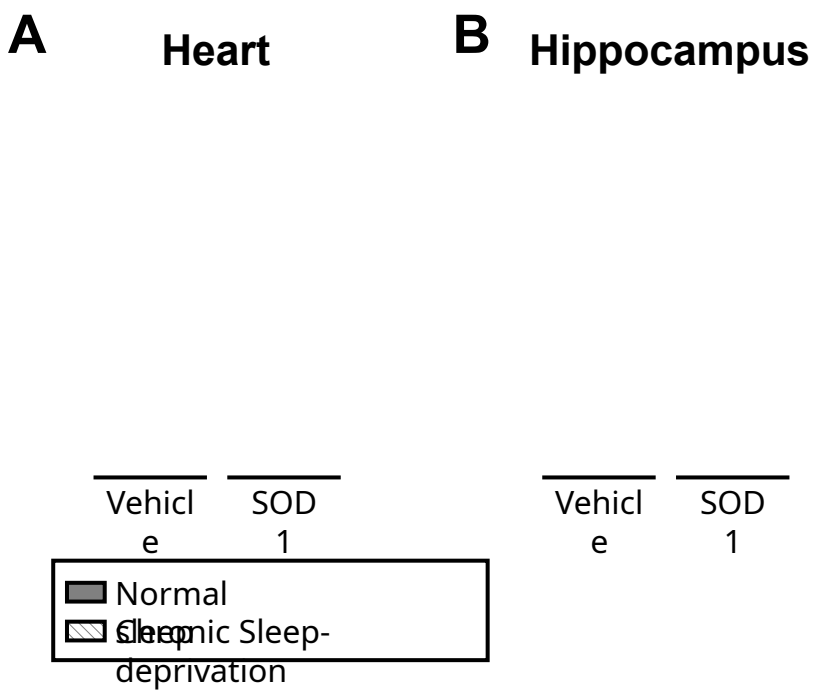


B



SOD1 assay

Lysis : SOD1 lysis buffer
→ 200uL of lysis buffer per 10mg of tissue
→ 1/50 dilute with SOD1 lysis buffer

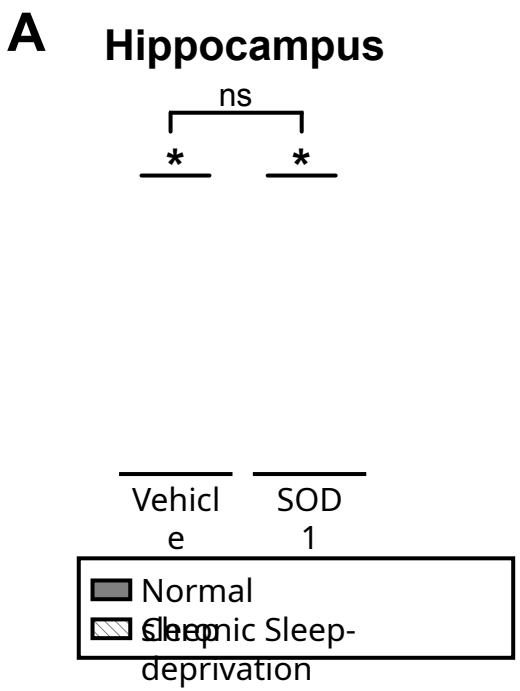


Effects of oral SOD1 supplementation and sleep deprivation on SOD activity in the heart and hippocampus.
Total superoxide dismutase 1 (SOD1) activity in (A) heart tissue and (B) hippocampal tissue from mice receiving SOD1 or vehicle (UV-sterilized water) in drinking water for 2 weeks, followed by 1 week chronic sleep deprivation (CSD) or normal sleep (NSL). Bars represent mean ± [SD/SEM]; each dot represents one mouse. Data were analyzed by two-way ANOVA, followed by Šidák's multiple comparisons test. *P < 0.05; ns, not significant.

Discussion
Sleep deprivation is associated with increased oxidative stress [1, 2], and the elevated cardiac SOD activity in vehicle-treated CSD mice likely reflects a compensatory upregulation of endogenous antioxidant defenses [3]. Notably, this response was absent in SOD1-supplemented mice (A). Because orally administered SOD is largely degraded in the gastrointestinal tract [6, 7], the measured activity most likely reflects endogenous enzyme rather than the supplement itself. One possible explanation is that SOD1 supplementation attenuated the oxidative burden associated with sleep deprivation, through indirect effects on gut or systemic redox status [2, 6], thereby reducing the stimulus for endogenous SOD induction [3]. This would be consistent with previous reports that exogenous antioxidants can blunt stress-induced upregulation of endogenous antioxidant enzymes [4, 5].
The absence of changes in the hippocampus suggests that the effects of oral SOD1 supplementation are tissue-specific. This may reflect limited systemic or central delivery of supplement-derived effects, or a lower responsiveness of hippocampal SOD activity to this sleep deprivation paradigm.
In contrast to the heart, hippocampal SOD activity was unaffected by SOD1 supplementation, sleep condition, or their interaction. This tissue specificity may be explained in part by the blood-brain barrier (BBB). SOD1 is a ~32 kDa protein, well above the size limit for passive diffusion across the BBB, and native SOD shows minimal brain penetration without specialized delivery strategies [8]. Thus, even if supplement-derived SOD or its effects reached the systemic circulation, they would be unlikely to directly influence the hippocampal redox environment. Furthermore, the BBB may buffer the brain from peripheral oxidative and inflammatory signals that could otherwise drive changes in endogenous SOD expression, so that central antioxidant

MDA assay

Lysis: 300uL of lysis buffer per 10mg of tissue



Sleep deprivation increases hippocampal lipid peroxidation independent of SOD1 supplementation.
Malondialdehyde (MDA) levels in hippocampal tissue from mice receiving SOD1 or vehicle (UV-sterilized water) in drinking water for 2 weeks, followed by 1 week of chronic sleep deprivation (CSD) or normal sleep (NSL).
Bars represent mean ± [SD/SEM]. Data were analyzed by two-way ANOVA, followed by Šidák's multiple comparisons test. *P < 0.05; ns, not significant.
*P < 0.05; ns, not significant.

Discussion
CSD significantly increased hippocampal MDA levels, a widely used marker of lipid peroxidation **[10]**, indicating that sleep deprivation induced oxidative damage in the hippocampus **[1]**. The brain is particularly susceptible to lipid peroxidation owing to its high oxygen consumption, abundance of polyunsaturated fatty acids, and relatively limited antioxidant capacity **[11]**. Notably, this increase in lipid peroxidation occurred without a corresponding change in hippocampal SOD activity, suggesting that the hippocampus did not upregulate SOD in response to CSD-induced oxidative stress. SOD1 supplementation affected neither hippocampal MDA levels nor SOD activity, consistent with limited access of supplement-derived SOD, or its downstream effects, to the brain across the blood-brain barrier **[8]**. Together, these findings indicate that oral SOD1 supplementation does not protect the hippocampus against sleep deprivation-induced lipid peroxidation.